Personal Expense Tracker

A Windows Desktop Application

Group Project Proposal
Group - 2
Academic Year 2024/2025 | Semester I

I. Project Name

Personal Expense Tracker

A Windows desktop application built with C# and .NET Framework that enables users to conveniently track their income, expenses, and budgets through an intuitive graphical interface backed by a Microsoft SQL Server database.

II. Aims and Objectives

Aim

To design and develop a comprehensive Personal Expense Tracker Windows application that helps individuals manage their finances effectively by recording income and expenses, setting budgets, and generating insightful financial reports.

Objectives

- Provide a secure user authentication system with login and registration functionality.
- Enable users to record and categorize income and expense transactions.
- Display a real-time financial dashboard summarizing income, expenses, and net balance.
- Allow users to set monthly budgets per category and monitor spending against those budgets.
- Generate visual financial reports (charts/graphs) to help users understand spending patterns.
- Maintain a complete transaction history with search and filter capabilities.
- Allow users to manage their personal profile information within the application.
- Ensure data persistence using Microsoft SQL Server as the backend database.

III. Brief Outline of the Proposed System

System Overview

The Personal Expense Tracker is a multi-form Windows Forms Application. Users first authenticate via the Login or Registration Form. Upon successful login, they are directed to the Dashboard Form which presents a summary of their finances. From there, users can navigate to any of the core functional forms. All data is stored and retrieved from a Microsoft SQL Server database using ADO.NET.

Windows Forms

#	Form Name	Description	Key Features
1	Login Form	Entry point for existing users to authenticate into the system.	Username/Password fields, Forgot Password link, Navigate to Register
2	Registration Form	Allows new users to create an account with personal details.	Full name, email, username, password, confirm password, validation
3	Dashboard Form	Main navigation hub showing financial summary at a glance.	Total income, total expenses, net balance, recent transactions, quick nav
4	Add Income Form	Form for recording new income entries with category and notes.	Amount, category, date picker, description, save/cancel
5	Add Expense Form	Form for recording new expense entries with category and notes.	Amount, category dropdown, date picker, description, save/cancel
6	Transaction History Form	Displays all past income and expense records with filtering options.	DataGridView, date range filter, category filter, search, delete/edit
7	Budget Planner Form	Allows users to set monthly spending limits per category.	Category budget setup, progress bars, overspend alerts
8	Reports Form	Generates visual financial summaries using charts and graphs.	Pie chart (expenses by category), bar chart (monthly trend), date filter
9	User Profile Form	Allows users to view and update their personal account details.	Name, email, password change, profile picture, save changes

UML Use Case Diagram

The following Use Case Diagram illustrates the interactions between the primary actor (User) and the system functionalities. The diagram covers all 11 identified use cases and shows the key include/extend relationships between them.

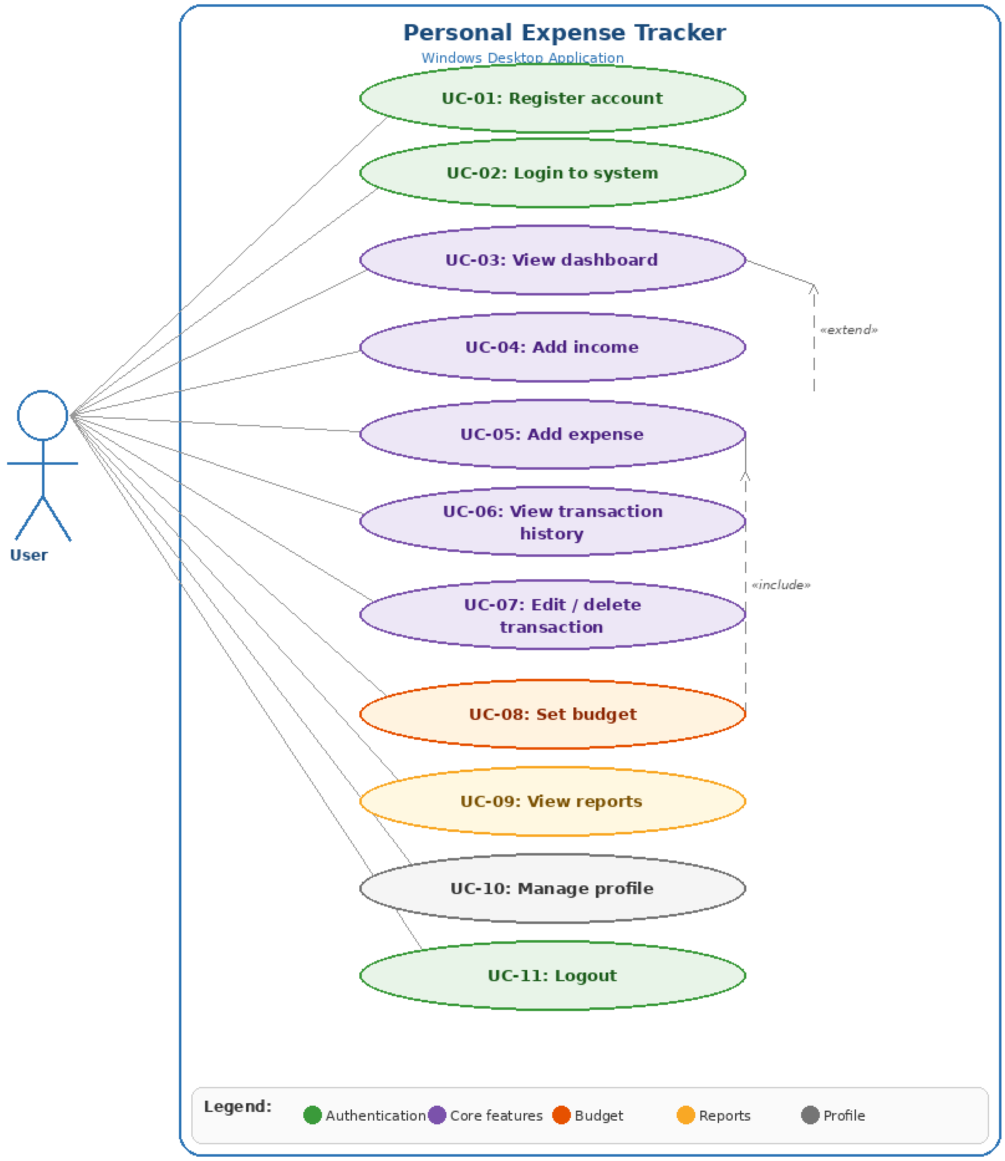

Figure 1: UML Use Case Diagram – Personal Expense Tracker

Use Cases Description

UC ID	Use Case	Description
UC-01	Register Account	A new user fills in their full name, email, username, and password to create an account. Includes input validation and duplicate username checks.
UC-02	Login to System	An existing user authenticates by entering valid credentials. Upon successful login the user is directed to the Dashboard.
UC-03	View Dashboard	The central hub showing real-time financial summary: total income, total expenses, net balance, and recent transactions. Extends after every transaction entry.
UC-04	Add Income	User records an income entry with amount, category, date, and optional description. Extends the Dashboard with updated figures.

UC ID	Use Case	Description
UC-05	Add Expense	User records an expense entry with amount, category, date, and optional description. Extends the Dashboard. Also included by UC-08.
UC-06	View Transaction History	Displays all income and expense records in a filterable, searchable DataGridView. Supports date range and category filters.
UC-07	Edit / Delete Transaction	User selects an existing transaction from history to modify its details or permanently remove it. Balance is recalculated accordingly.
UC-08	Set Budget	User configures monthly spending limits per category. Includes UC-05 to track expenses against set budgets in real time with progress bars and alerts.
UC-09	View Reports	Generates visual charts: a pie chart for expense distribution by category and a bar chart for monthly income/expense trends with date range filtering.
UC-10	Manage Profile	User views and updates personal account details including name, email, password, and profile picture.
UC-11	Logout	Ends the current user session and redirects the user back to the Login Form.

UML Relationships

- Include relationship: UC-08 (Set Budget) includes UC-05 (Add Expense) — the budget planner integrates real-time expense recording to compare spending against defined limits.
- Extend relationships: UC-04 (Add Income) and UC-05 (Add Expense) extend UC-03 (View Dashboard) — the dashboard totals are automatically refreshed after each transaction entry.

Database Design Overview

The Microsoft SQL Server database will consist of the following primary tables:

Table	Columns
Users	UserID (PK), FullName, Username, Email, PasswordHash, ProfilePicture, CreatedAt
Transactions	TransactionID (PK), UserID (FK), Type (Income/Expense), Amount, CategoryID (FK), Description, Date
Categories	CategoryID (PK), CategoryName, Type (Income/Expense)
Budgets	BudgetID (PK), UserID (FK), CategoryID (FK), MonthYear, BudgetAmount

IV. Tools and Technologies

Technology / Tool	Purpose
C# Programming Language	Core application logic, event handling, and data processing
.NET Framework	Application framework; Windows Forms GUI support
Visual Studio IDE	Development environment, form designer, and debugging
Microsoft SQL Server (SSMS)	Relational database for persistent data storage
ADO.NET	Database connectivity, SQL queries, and data binding
Windows Forms (WinForms)	GUI design — forms, controls, event-driven programming
DataGridView Control	Displaying transaction history in tabular format
Chart Control (MS Chart)	Generating pie charts and bar charts in the Reports Form
Git / GitHub	Version control and team collaboration

V. Initial Plan and Work Allocation

Project Timeline

Phase	Activity	Target Date
Phase 1	Project Proposal Submission	30th April 2026
Phase 2	Database Design & Setup	1st – 7th May 2026
Phase 3	UI Design & Individual Form Development	8th – 25th May 2026
Phase 4	Integration & Testing	26th – 31st May 2026
Phase 5	Final Review & Submission	June 2026

Work Allocation

#	Member Name	Student No.	Assigned Forms	Additional Responsibilities
1	T.M.D.P.M De Alwis	EC/2022/012	Login Form, Registration Form	Project coordination, DB setup, Authentication logic
2	P.M.R.N.D.B Wathurakumbura	EC/2022/056	Dashboard Form, User Profile Form	UI/UX design consistency, Navigation logic
3	K.P Tharuka	PS/2022/205	Add Income Form, Add Expense Form	Transaction data models, Category management
4	W.P.M Manavi	EC/2022/062	Transaction History Form, Budget Planner Form, Reports Form	Reporting & charts, SQL queries, Testing

VI. Group Member Details

#	Full Name (with Initials)	Student Number	Student Email	Role
1	T.M.D.P.M De Alwis	EC/2022/012	dealwis-ec22012@stu.kln.ac.lk	Group Leader
2	W.P.M Manavi	EC/2022/062	manaviw-ec22062@stu.kln.ac.lk	Member
3	P.M.R.N.D.B Wathurakumbura	EC/2022/056	wathura-ec22056@stu.kln.ac.lk	Member
4	K.P Tharuka	PS/2022/205	tharuka-ps22205@stu.kln.ac.lk	Member